TREVOR TAYLOR

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EDUCATION

Georgia Institute of Technology | M.S. Aerospace Engineering 2025 – 2027
San Diego State University | B.S. Aerospace Engineering 2020 – 2024

SKILLS

- Hypermesh
- Patran
- NASTRAN
- CATIA V5
- MATLAB
- Teamcenter
- Python / C / VBA
- Configuration Management
- SolidWorks

EXPERIENCE

Structural Stress Engineer II | Collins Aerospace, Aerostructures FEBRUARY 2025 – PRESENT

Stress analysis for the Boeing 787, Airbus A220 / Embraer E2 aircraft nacelles, RTX missiles, and tooling

- Performing structural analysis of primary and secondary nacelle structures using Hypermesh, NASTRAN, Patran, and hand calculations to determine margins under static, dynamic, and fatigue loads
- Analyzed several tooling structures to support the production of various missile systems by conceptualizing load paths, performing stress analyses, making necessary design/material changes, and writing full analysis records
- Utilizing validated mitigation techniques to correct negative margin issues in finite element models and associated analysis
- Engaged with suppliers on corrective actions for non-compliant parts and completed one-off structural assessments to validate continued safety and flight readiness
- Created Excel macros and Python scripts to automate data extraction, analysis, and reporting
- Developed and revised engineering standard operating manuals detailing methodologies for bolted joint and weld analysis
- Led material trade studies for cracked secondary structures, improving reliability and preventing continued field issues

Mechanical Engineer I | Collins Aerospace, Aerostructures JUNE 2024 – FEBRUARY 2025

Led development of lifecycle processes for engineering documentation across teams during a PLM migration for ~2,000 users

- Developed tailored process solutions to meet individual program needs for Airbus, Boeing, and Gulfstream aircraft

PROJECTS

TVC Rocket + Launch Pad | Personal Project MAY 2026 – PRESENT

Designed and built an actively stabilized, thrust-vectored rocket and integrated launch system

- Designed a custom STM32 flight computer with an IMU, baro & flash memory for real-time sensing, PD thrust vector control & data logging
- Implemented a Mahony attitude filter and a 3-state Kalman filter in bare-metal C, fusing IMU and barometric data at 200Hz to estimate orientation, altitude & vertical velocity
- Developed a flight state machine with sensor-based launch, burnout & apogee detection, driving a custom, spring-loaded mechanical parachute ejection system
- Built a Python post-flight analysis tool decoding binary flash logs into plots and 3D attitude replay; used flight data to diagnose control saturation and identify root causes
- Built a launch pad with custom computer for remote ignition, servo-actuated tower arm retraction & pad status indication

Senior Design Lead, Solid Rocket Motors | San Diego State University JANUARY 2024 – MAY 2024

Directed a team of five students in analyzing the performance of various propellant grain geometries in class K solid rocket motors

- Performed internal ballistics calculations to predict burn rate, thrust, chamber pressure, and other relevant parameters
- Led the development of two solid rocket motors, from inception to testing and comprehensive analysis
- Designed and constructed a test stand featuring embedded data acquisition for measuring thrust over time